Example 8.28 If A and B are two events such that $P(A) = \frac{1}{4}$, $P(B) = \frac{1}{2}$ and $P(A \text{ and } B) = \frac{1}{8}$, find (i) $P(A \text{ or } B)$ (ii) $P(\text{not } A \text{ and not } B)$.

Solution (i) $P(A \text{ or } B) = P(A \cup B)$

$$= P(A) + P(B) - P(A \cap B)$$

$$P(A \text{ or } B) = \frac{1}{4} + \frac{1}{2} - \frac{1}{8} = \frac{5}{8}$$

(ii) $P(\text{not } A \text{ and not } B) = P(\bar{A} \cap \bar{B})$

$$= P\left(\overline{A \cup B}\right)$$

$$= 1 - P\left(A \cup B\right)$$

$$P(\text{not } A \text{ and not } B) = 1 - \frac{5}{8} = \frac{3}{8}$$

Example 8.29 In an apartment, in selecting a house from door numbers 1 to 100 randomly, find the probability of getting the door number of the house to be an even number or a perfect square number or a perfect cube number

Solution:

Total number of houses $n(S) = 100$

Let A be the event of getting door number even.

$$\text{A} = \{2, 4, 6, 8 \ldots\ldots\ldots 100\}$$

$$n(A) = 50$$

$$P(A) = \frac{n(A)}{n(S)} = \frac{50}{100}$$

Let B be the event of getting door number perfect square.

$$\text{B} = \{1, 4, 9, 16, 25, 36, 49, 64, 81, 100\}$$

$$\therefore n(B) = 10$$

$$P(B) = \frac{n(B)}{n(S)} = \frac{10}{100}$$

Let C be the event of getting door number perfect cube.

$$\text{C} = \{1, 8, 27, 64\}$$

$$\therefore n(C) = 4$$

$$P(C) = \frac{n(C)}{n(S)} = \frac{4}{100}$$

$P(A \cap B) = P$ [getting even perfect square number] $= \frac{5}{100}$

$P(B \cap C) = P$ [getting a perfect square and perfect cube number] $= \frac{2}{100}$

$P(A \cap C) = P$ [getting even perfect cube number] $= \frac{2}{100}$

$P(A \cap B \cap C) = P$ [getting even perfect square and perfect cube number] $= \frac{1}{100}$

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